Dhirubhai Ambani University

(Formerly known as DA-IICT)

Topic: GUI in Python-Tkinter

Dr. Ankit Vijayvargiya

Assistant Professor

Room No. 4205, Faculty Block 4

Email: ankit_Vijayvargiya[at]dau.ac.in

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GUI

What is GUI?

- A Graphical user interface (GUI) is an application that features buttons, windows, and various other widgets that the user can use to interact with the application.
- A good example would be a web browser. It features buttons, tabs, and a main window where all the content is displayed.

Why is a GUI Needed?

- **User-Friendly:** Intuitive for non-technical users.
- **Efficiency:** Faster interactions via point-and-click.
- **Visual Feedback:** Immediate response to actions.
- **Multitasking:** Supports multiple tasks via windows/menus.
- **Accessibility:** Easier for diverse users, including those with disabilities.

GUI

Python Libraries for GUI Development:

- Tkinter: Built-in, lightweight, cross-platform, beginner-friendly.
- PyQt: Qt-based, ideal for complex, professional apps.
- Kivy: Open-source, multi-touch apps for mobile/desktop.
- wxPython: Native look-and-feel, cross-platform.
- PyGTK: GTK-based, suited for GNOME environments.
- PySide: Qt-based, free for commercial use.
- BeeWare: Native apps for multiple platforms.

Choosing a library depends on the project's complexity, platform, and the features required.

Why Tkinter

Tkinter: Python's standard GUI library, included with Python.

Key Benefits:

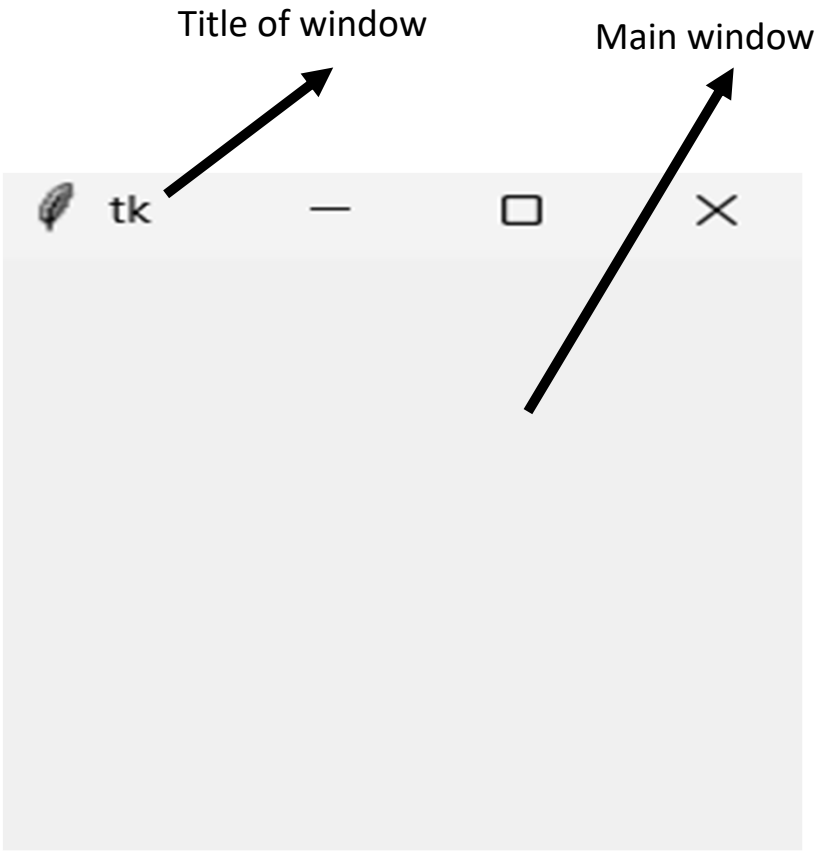
- Built-in, no extra installation.
- Simple and intuitive for beginners.
- Cross-platform (Windows, macOS, Linux).
- Event-driven programming for responsive apps.
- Wide range of widgets for calculators, forms, etc.

Now, let us move further with basic tkinter setup and creation of labels, widgets, etc.

First Tkinter Application

Steps to Create a Tkinter App:

- Import the tkinter module.
- Create the main applicationwindow with Tk().
- After creating the main window, add the widgets (e.g., buttons, labels, frames).
- Call the main event loop with mainloop(), so that the action can take place on the user's computer screen.



Code Example:

```
import tkinter as tk
from tkinter import ttk #widget
```

```
window = tk.Tk()

#Add widgets button over here

window.mainloop()
```

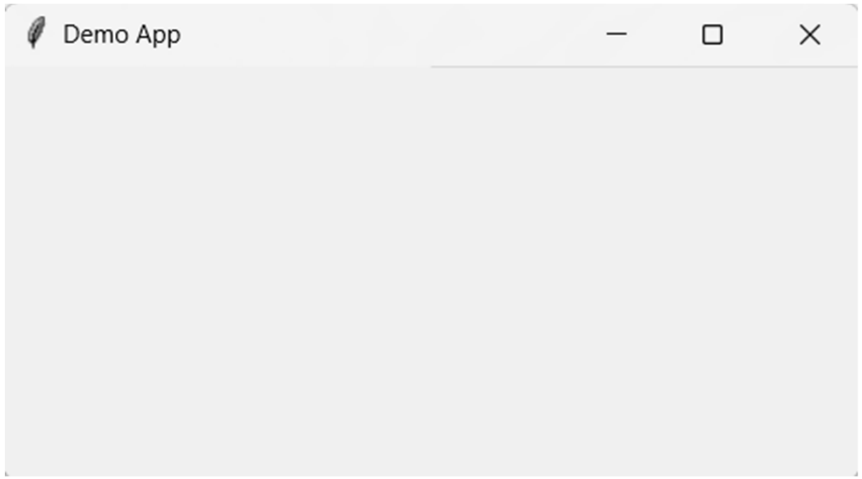
GUI

Add Title

```
window = tk.Tk()


window.title("Demo App")

window.mainloop()
```

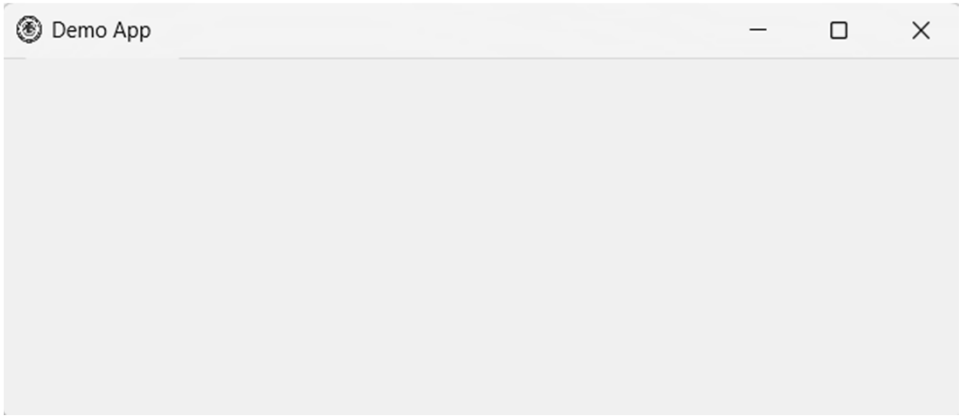


Change Icon

```
window = tk.Tk()
window.title("Demo App")

icon = tk.PhotoImage(file=r"D:/1111.png")
window.iconphoto(False, icon)

window.mainloop()
```



GUI

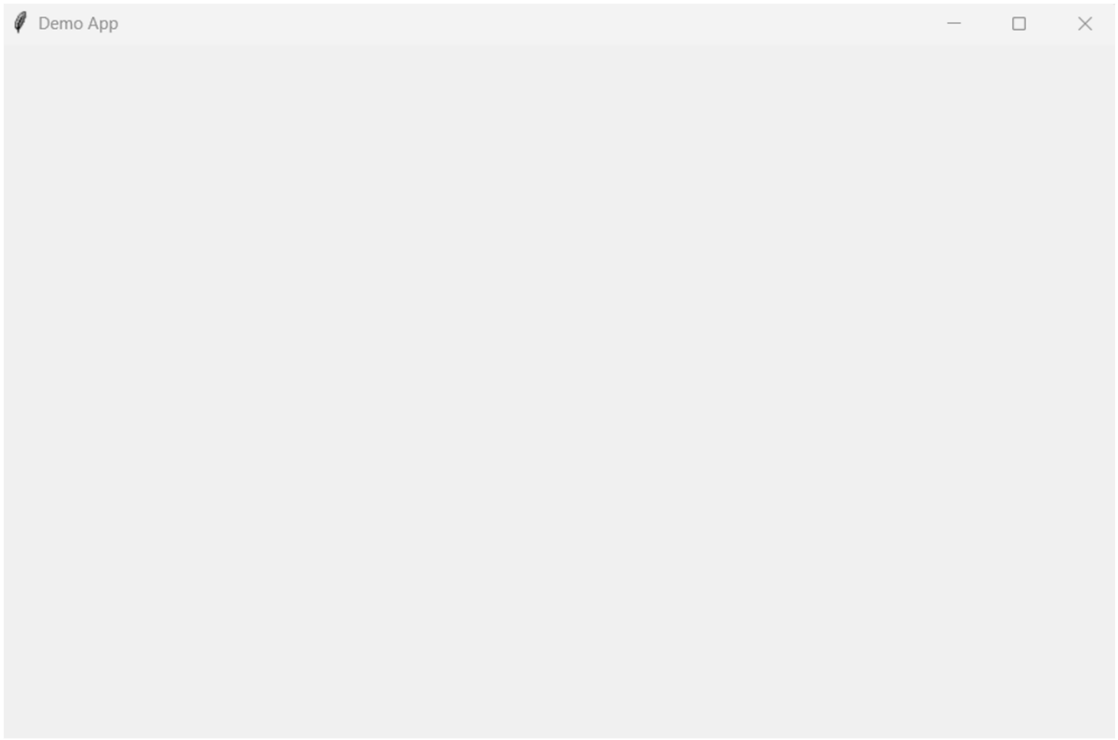
Size of UI

```
window = tk.Tk()

window.title("Demo App")

window.geometry("800x500") # size (Pixel)

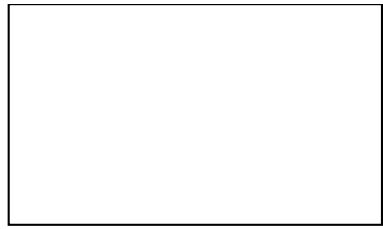
window.mainloop()
```



GUI

Tkinter widgets or components

Widget	Description
Button	Creates various buttons in Python Application.
Checkbutton	Select one or more options from multiple options.(Checkbox)
Entry	Allows the user to enter single line of text(Textbox)
Frame	Acts like a container which can be used to hold the other widgets
Label	Used to display non editable text on window
Listbox	Display the list items, The user can choose one or more items.
Radiobutton	Select one option from multiple options.
Text	Allows the user to enter single or multiple line of text(Textarea)
Scale	Creates the graphical slider, the user can slide through the range of values
Toplevel	Used to create and display the top-level windows(Open a new window)



Label

GUI

Tkinter - Widget- Label

- **A label** is a widget used to **display text or images** on the screen.
- It is a *non-interactive* element, meaning the user **cannot type or click** inside it (it's only for showing information).

```
from tkinter import ttk #widget
```

```
window = tk.Tk()
window.title("Demo App")
window.geometry("800x500") # size (Pixel)

heading=ttk.Label(window,text="Hello World")
heading.pack()

window.mainloop()
```



GUI

Layout: Pack/Grid/Place

Grid Method:

- **The grid() method organizes the widgets in a tabular form. We can specify the rows and columns in the method call.**

Syntax:

widget.grid(options)

The possible options are:

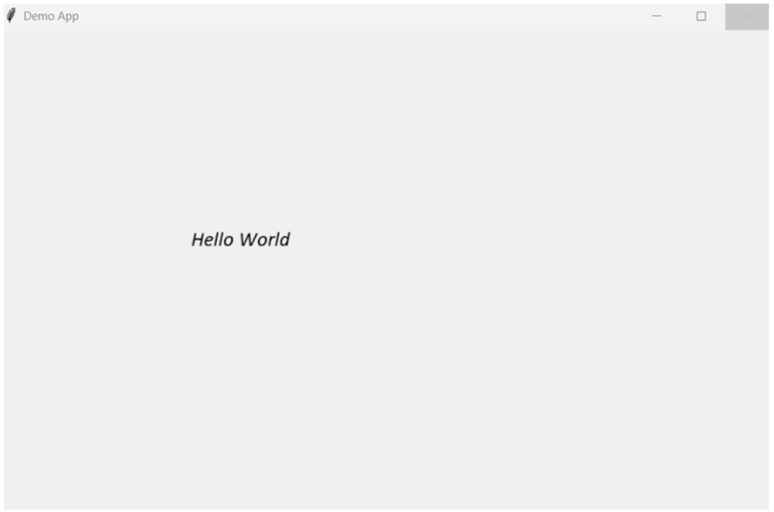
- **Column:** Column number in which the widget to be placed
- **Padx, pady:** represent the number of pixels to pad the widget outside the widget's border
- **Row:** row number in which the widget is to be placed

```
import tkinter as tk
from tkinter import ttk

window = tk.Tk()
window.title("Demo App")
window.geometry("800x500")  # size (Pixel)

heading = ttk.Label(master=window, text="Hello World",
                    font="calibri 16 italic")
heading.grid(row=20, column=0, padx=20, pady=20)

window.mainloop()
```



GUI

Layout: Pack/Grid/Place

Pack Method:

- **The pack() method is used to organise components or widgets in the main window.**

Syntax:

widget.pack(options)

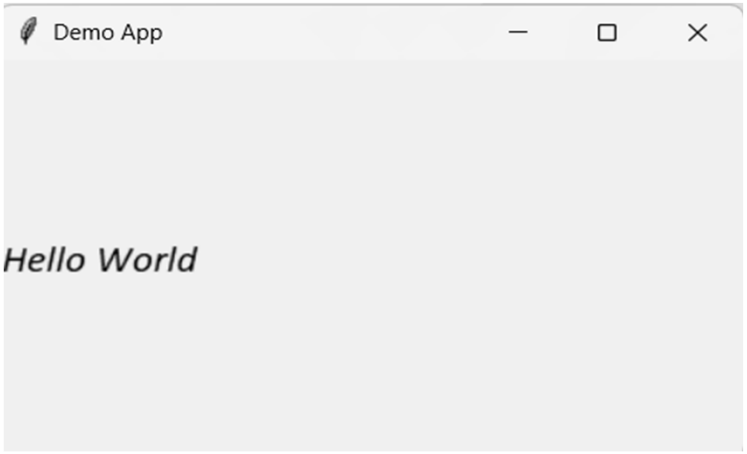
The possible options are:

- **side:** it represents the side to which the widget is to be placed on the window. Side may be Left, right, top (default) or bottom.

```
window = tk.Tk()
window.title("Demo App")
window.geometry("800x500") # size (Pixel)

heading=ttk.Label(master=window,text="Hello World",
                  font="calibri 16 italic")
heading.pack(side="left")

window.mainloop()
```



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Layout

Place Method:

- **The place() method organizes the widgets to the specific x and y coordinates.**

Syntax:

widget.place(x,y)

- **x,y:** refer to the horizontal and vertical offset in pixels.

```
import tkinter as tk
from tkinter import ttk

window = tk.Tk()
window.title("Demo App")
window.geometry("800x500")  # size (Pixel)

heading = ttk.Label(master=window, text="Hello World",
                    font="calibri 16 italic")
heading.place(x=500, y=300)

window.mainloop()
```



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GUI

Tkinter - Widget- Label

```
window = tk.Tk()
window.title("Demo App")
window.geometry("800x500") # size (Pixel)

heading=ttk.Label(master=window,text="Hello World",
                  font="calibri 16 italic",background="red", foreground="yellow")
heading.pack()

window.mainloop()
```



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GUI

Tkinter - Widget- Label- Image

```
import tkinter as tk
from tkinter import ttk

window = tk.Tk()
window.title("Image Example")
window.geometry("400x400")

# Load image (PNG/GIF only)
img = tk.PhotoImage(file=r"D:/1111.png")

# Make a Label to show image
label = ttk.Label(window, image=img)
label.pack()

window.mainloop()
```



GUI

Tkinter - Widget- Label- Image- Size

```
import tkinter as tk
from tkinter import ttk

window = tk.Tk()
window.title("Image Example")
window.geometry("400x400")

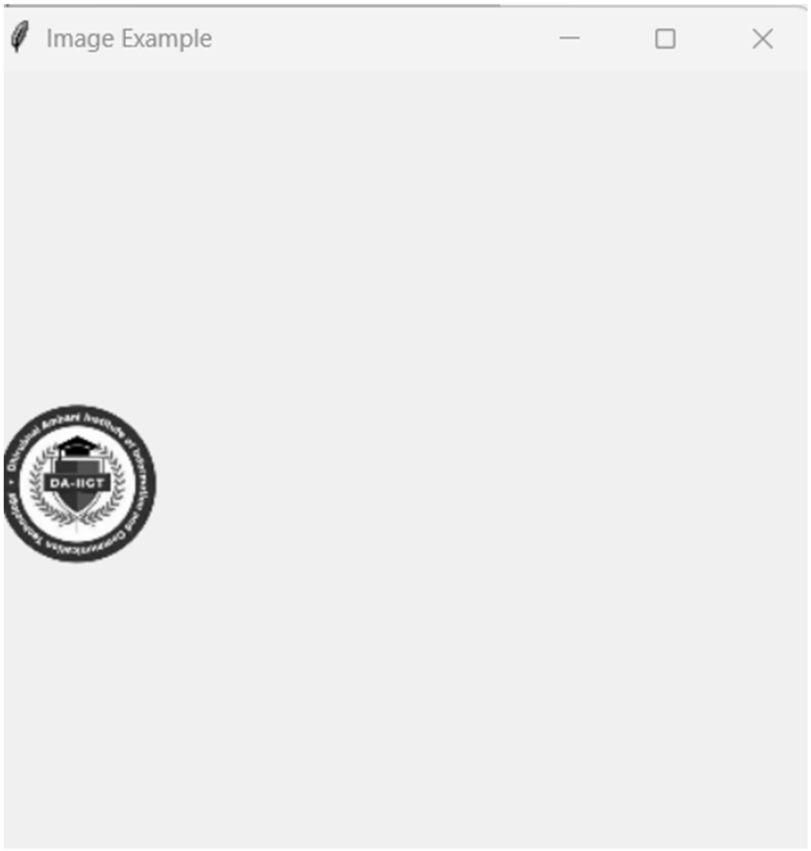
# Load image (PNG/GIF only)
img = tk.PhotoImage(file=r"D:/1111.png")

# to make it smaller
img = img.subsample(3, 3)

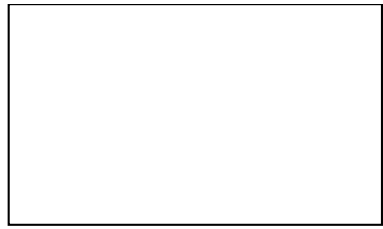
#to make it larger
#img = img.zoom(1, 1)

# Make a Label to show image
label = ttk.Label(window, image=img)
label.pack(side="left")

window.mainloop()
```



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Entry

Tkinter - Widget- Entry

- The entry widget is used to provide a single-line text box to the user to accept a value from them.
- We can use an entry widget to accept a text string from the user.

Syntax:

name= Entry(parent, options)

GUI

Tkinter - Widget- Entry

```
window = tk.Tk()
window.title("Demo App")
window.geometry("800x500") # size (Pixel)


heading_1=ttk.Label(window,text="Name",font="calibri 16 italic")

inputBox=ttk.Entry(window)

heading_1.pack()
inputBox.pack()

window.mainloop()
```



GUI

Tkinter - Widget- Entry

```
window = tk.Tk()
window.title("Demo App")
window.geometry("800x500") # size (Pixel)


heading_1=ttk.Label(window,text="Name",font="calibri 16 italic")

inputBox=ttk.Entry(window, width=50, foreground="red", font="calibri 16 italic")

heading_1.pack()
inputBox.pack()

window.mainloop()
```



GUI

Tkinter - Widget- Entry

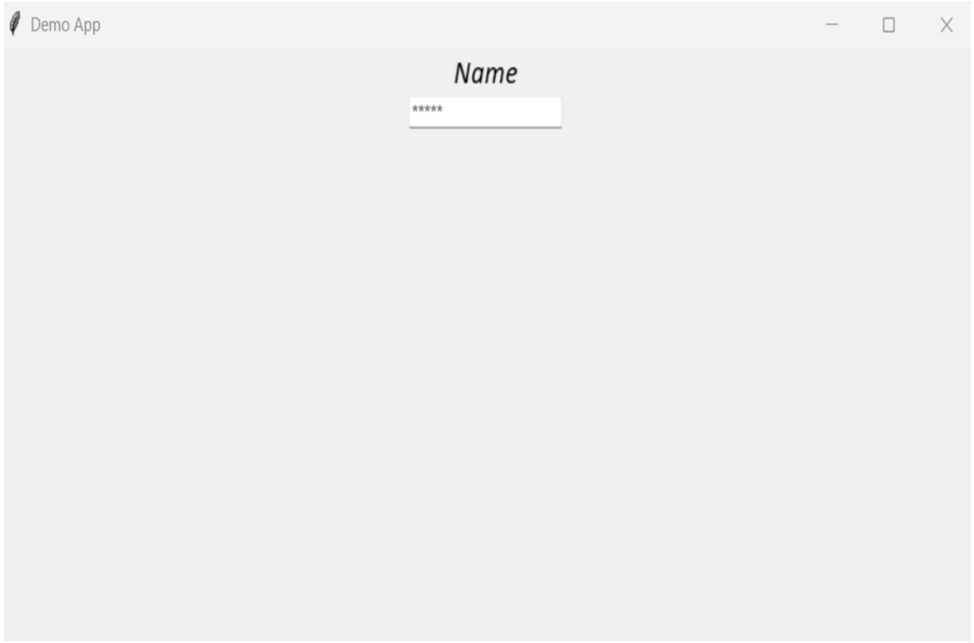
```
window = tk.Tk()
window.title("Demo App")
window.geometry("800x500") # size (Pixel)


heading_1=ttk.Label(window,text="Name",font="calibri 16 italic")

inputBox=ttk.Entry(window, show="*", foreground="red")

heading_1.pack()
inputBox.pack()

window.mainloop()
```



GUI

Tkinter - Style

```
window = tk.Tk()
window.title("Demo App")
window.geometry("800x500") # size (Pixel)

style = ttk.Style()
style.configure("TLabel", font="calibri 24 italic",
                background="red", foreground="yellow")

heading=ttk.Label(master=window,text="Hello World", style="TLabel")

heading.pack()

window.mainloop()
```

TFrame	ttk.Frame
TLabel	ttk.Label
TButton	ttk.Button
TEntry	ttk.Entry
TCheckbutton	ttk.Checkbutton
TRadiobutton	ttk.Radiobutton
TCombobox	ttk.Combobox
TNotebook	ttk.Notebook
TProgressbar	ttk.Progressbar
TScale	ttk.Scale
TScrollbar	ttk.Scrollbar
TSpinbox	ttk.Spinbox
Treeview	ttk.Treeview

Tkinter - Style

```
window = tk.Tk()
window.title("Demo App")
window.geometry("800x500") # size (Pixel)

style = ttk.Style()
style.configure("h.TLabel", font="calibri 24 italic",
                background="red", foreground="yellow")

style.configure("h1.TLabel", font="calibri 24 italic",
                background="green", foreground="blue")

heading=ttk.Label(master=window,text="Hello World", style="h.TLabel")

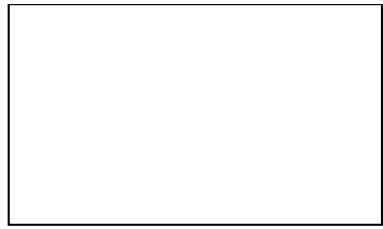
heading_1=ttk.Label(master=window,text="How Are YOU", style="h1.TLabel")

heading.pack()

heading_1.pack()

window.mainloop()
```





Frames

Tkinter - Widget- Frame

- The frame widget is used to organise a group of widgets. It acts like a container that can be used to hold the other widgets.
- The rectangular area of the screen is used to organise the widgets in the Python applications.

Syntax:

name= Frame(parent, options)

GUI

Tkinter - Widget- Frame

```
import tkinter as tk
from tkinter import ttk


window = tk.Tk()
window.title("Demo App")
window.geometry("500x300") # size (Pixel)

style = ttk.Style()
style.configure("Red1.TFrame", background="red")
Frame_1 = ttk.Frame(window, style="Red1.TFrame")

style.configure("Red.TFrame", background="yellow")
Frame_2 = ttk.Frame(window, style="Red.TFrame")

heading_1=ttk.Label(Frame_1,text="First Name",font="calibri 16 italic")
inputBox_1=ttk.Entry(Frame_1)

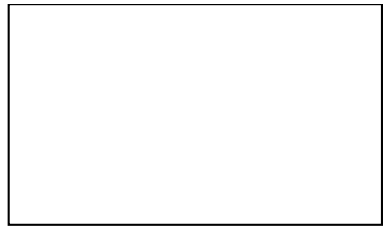
heading_1.pack(side="left", padx=20)
inputBox_1.pack(side="left", padx=20)


heading_2=ttk.Label(Frame_2,text="Last Name",font="calibri 16 italic")
inputBox_2=ttk.Entry(Frame_2)

heading_2.pack(side="left", padx=20)
inputBox_2.pack(side="left", padx=20)
Frame_1.pack()
Frame_2.pack()

window.mainloop()
```





Buttons

Tkinter - Widget- Button

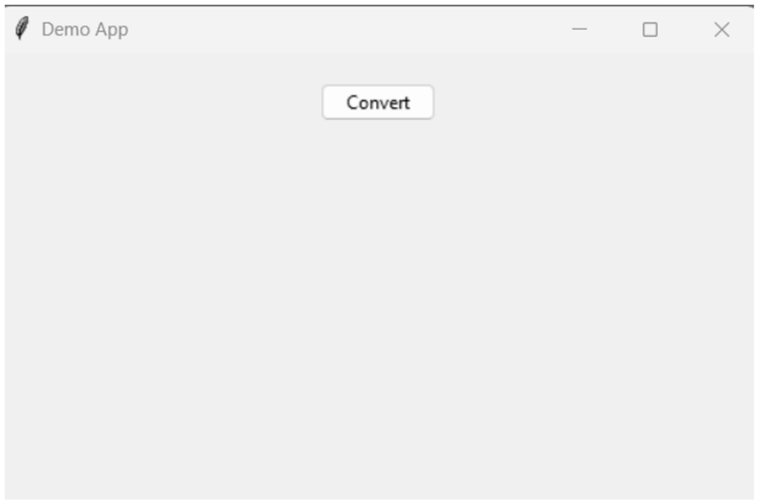
- The button is used to add various kinds of buttons to the python application. We can also associate a method or a function with a button which is called when the button is pressed.

- Syntax: name = Button(parent, options)

```
window = tk.Tk()
window.title("Demo App")
window.geometry("500x300") # size (Pixel)

button=ttk.Button(master=window, text="Convert")
button.pack(pady=20)

window.mainloop()
```



Tkinter - Widget- Button

Normal state style

```
window = tk.Tk()
window.title("Demo App")
window.geometry("500x300") # size (Pixel)

style = ttk.Style()

# Normal state style
style.configure("TButton",background="red",
    foreground="green",
    font=("Calibri", 12, "bold"),
    padding=5,
    borderwidth=2)

button=ttk.Button(master=window, text="Convert",style="TButton")
button.pack(pady=20)

window.mainloop()
```

Hover/active style

```
window = tk.Tk()
window.title("Demo App")
window.geometry("500x300") # size (Pixel)

style = ttk.Style()


# Hover / active style
style.map(
    "TButton",
    background=[("active", "blue")],
    foreground=[("active", "yellow")]
)
button=ttk.Button(master=window, text="Convert",style="TButton")
button.pack(pady=20)

window.mainloop()
```



GUI

Tkinter - Widget- Button- Call Function

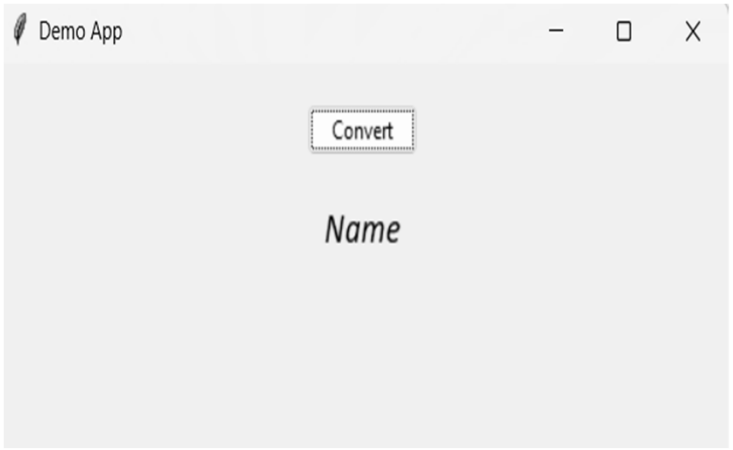
```
window = tk.Tk()
window.title("Demo App")
window.geometry("500x300") # size (Pixel)

def function():
    heading_1=ttk.Label(window,text="Name",font="calibri 16 italic")
    heading_1.pack()

button=ttk.Button(master=window, text="Convert", command=function)
button.pack(pady=20)

window.mainloop()
```





Tkinter - Widget- Checkbutton

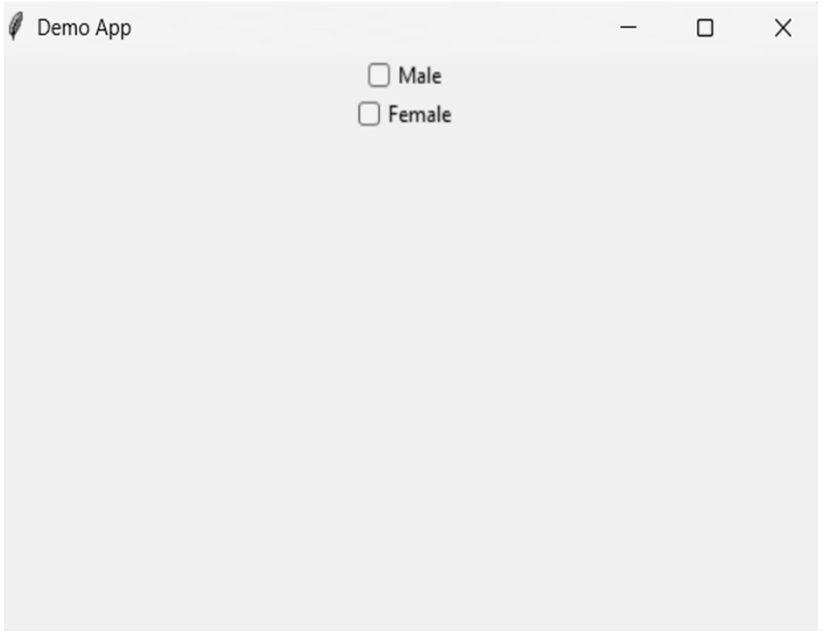
Checkbutton Widget: Toggles on/off states, linked to a variable.

```
window = tk.Tk()
window.title("Demo App")
window.geometry("800x500")  # size (Pixel)

gender_var_1 = tk.BooleanVar()
gender_var_2 = tk.BooleanVar()
male_radio = ttk.Checkbutton(window, text="Male", variable=gender_var_1)
female_radio = ttk.Checkbutton(window, text="Female", variable=gender_var_2)


male_radio.pack()
female_radio.pack()

window.mainloop()
```



Tkinter - Widget- Checkbutton

Checkbutton Widget: Toggles on/off states, linked to a variable.

```
import tkinter as tk # logical operator
from tkinter import ttk #widget

window = tk.Tk()
window.title("Demo App")
window.geometry("800x500")  # size (Pixel)

gender_var_1 = tk.BooleanVar(value="True")
gender_var_2 = tk.BooleanVar()
male_check = ttk.Checkbutton(window, text="Male", variable=gender_var_1)
female_check = ttk.Checkbutton(window, text="Female", variable=gender_var_2)


male_check.pack()
female_check.pack()

window.mainloop()
```



Tkinter - Widget- Radiobutton

Radiobutton Widget: Allows selecting one option from a group, sharing a variable.

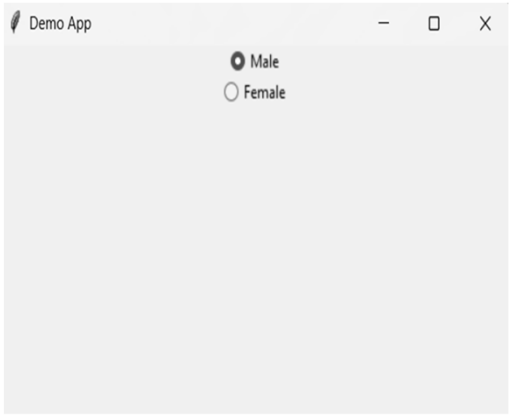
```
window = tk.Tk()
window.title("Demo App")
window.geometry("800x500")  # size (Pixel)


# Use ONE variable for both options
gender_var = tk.StringVar(value="Male")  # default empty

male_radio = ttk.Radiobutton(window, text="Male", variable=gender_var, value="Male")
female_radio = ttk.Radiobutton(window, text="Female", variable=gender_var, value="Female")


male_radio.pack()
female_radio.pack()

window.mainloop()
```




```
window = tk.Tk()
window.title("Demo App")
window.geometry("800x500")

gender_var = tk.IntVar(value=1)

male_radio = ttk.Radiobutton(
    window,
    text="Male",
    variable=gender_var,
    value=1
)

female_radio = ttk.Radiobutton(
    window,
    text="Female",
    variable=gender_var,
    value=2
)

male_radio.pack()
female_radio.pack()

window.mainloop()
```

```
print(gender_var.get())
```

```
Female
```

Tkinter - Widget- Dropdown

```
import tkinter as tk
from tkinter import ttk

window = tk.Tk()
window.title("Demo App")
window.geometry("500x300")  # size (Pixel)

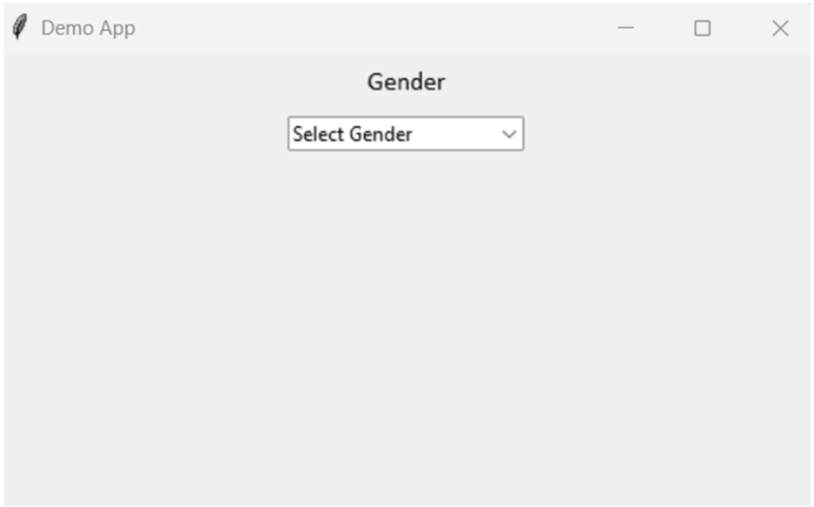

# Dropdown menu for Gender
gender_var = tk.StringVar()

gender_label = ttk.Label(window, text="Gender", font="calibri 12")

gender_dropdown = ttk.Combobox(window, textvariable=gender_var)
gender_dropdown['values'] = ("Male", "Female", "Other", "Other_1")
gender_dropdown.set("Select Gender")  # default display text


gender_label.pack(pady=5)
gender_dropdown.pack(pady=5)


window.mainloop()
```



Tkinter - Widget- Listbox

```
import tkinter as tk
from tkinter import ttk

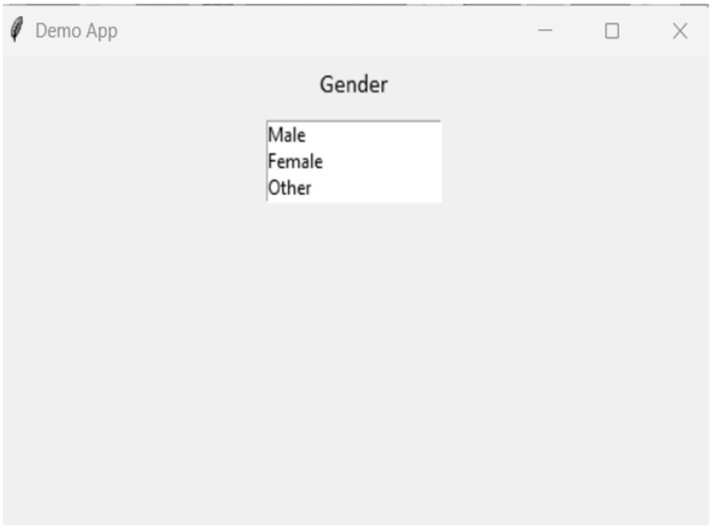
window = tk.Tk()
window.title("Demo App")
window.geometry("500x300")  # size (Pixel)

# Listbox for Gender
gender_label = ttk.Label(window, text="Gender", font="calibri 12")

gender_listbox = tk.Listbox(window, height=3)  # height=number of visible items, selectmode=tk.MULTIPLE
gender_listbox.insert(1, "Male")
gender_listbox.insert(2, "Female")
gender_listbox.insert(3, "Other")

gender_label.pack(pady=5)
gender_listbox.pack(pady=5)


window.mainloop()
```



GUI

Tkinter - Widget- Slider

```
import tkinter as tk
from tkinter import ttk

window = tk.Tk()
window.title("Demo App")
window.geometry("500x300")  # size (Pixel)


# Scale (slider)
scale_label = ttk.Label(window, text="Select Age", font="calibri 12")

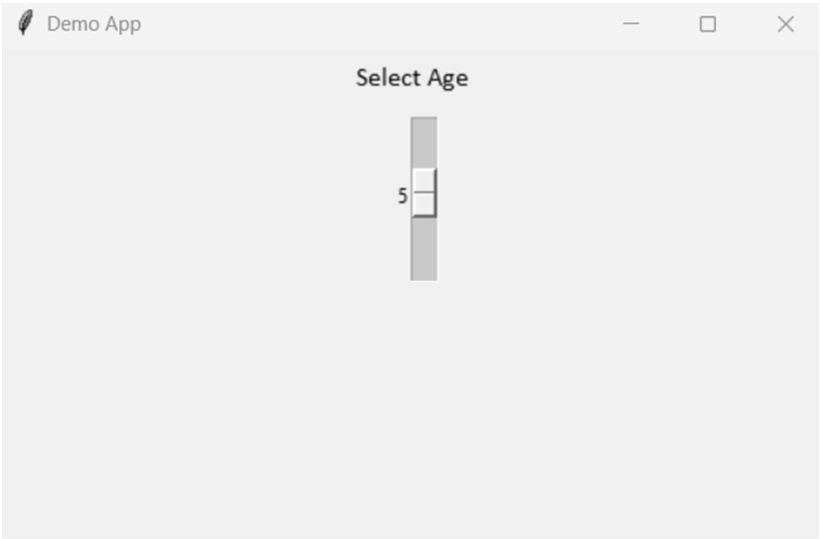
age_var = tk.IntVar(value=5)  # default value

age_scale = tk.Scale(window, from_=1,to=10, orient="vertical", variable=age_var)

#vertical orientation
#age_scale = tk.Scale(window, from_=1,to=100, orient="vertical", variable=age_var)

scale_label.pack(pady=5)
age_scale.pack(pady=5)


window.mainloop()
```



GUI

Tkinter - Widget- Multiline Text Box

```
window = tk.Tk()
window.title("Demo App")
window.geometry("500x300")  # size (Pixel)


heading_1 = ttk.Label(master=window, text="Name / Description", font="calibri 16 italic")

# Multiline Text Box
inputBox = tk.Text(window, width=40, height=5)  # width=chars, height=lines

heading_1.pack(padx=20, pady=5)
inputBox.pack(pady=5)

window.mainloop()
```



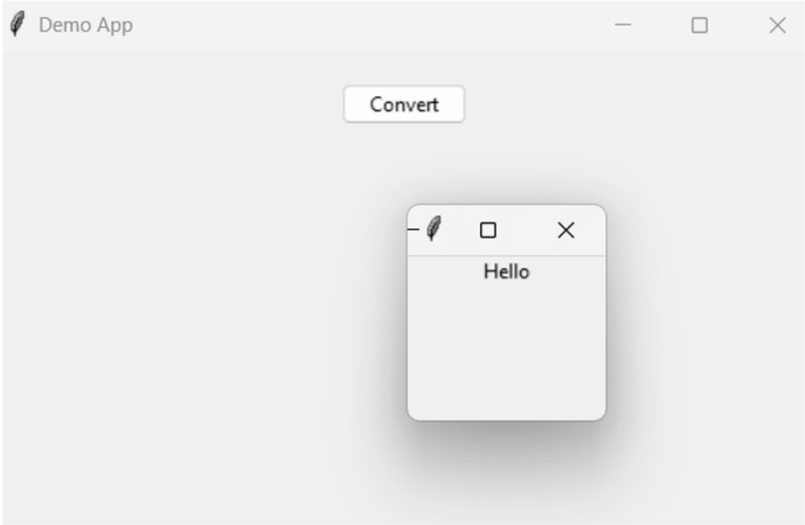
Tkinter – New Window

```
window = tk.Tk()
window.title("Demo App")
window.geometry("500x300") # size (Pixel)

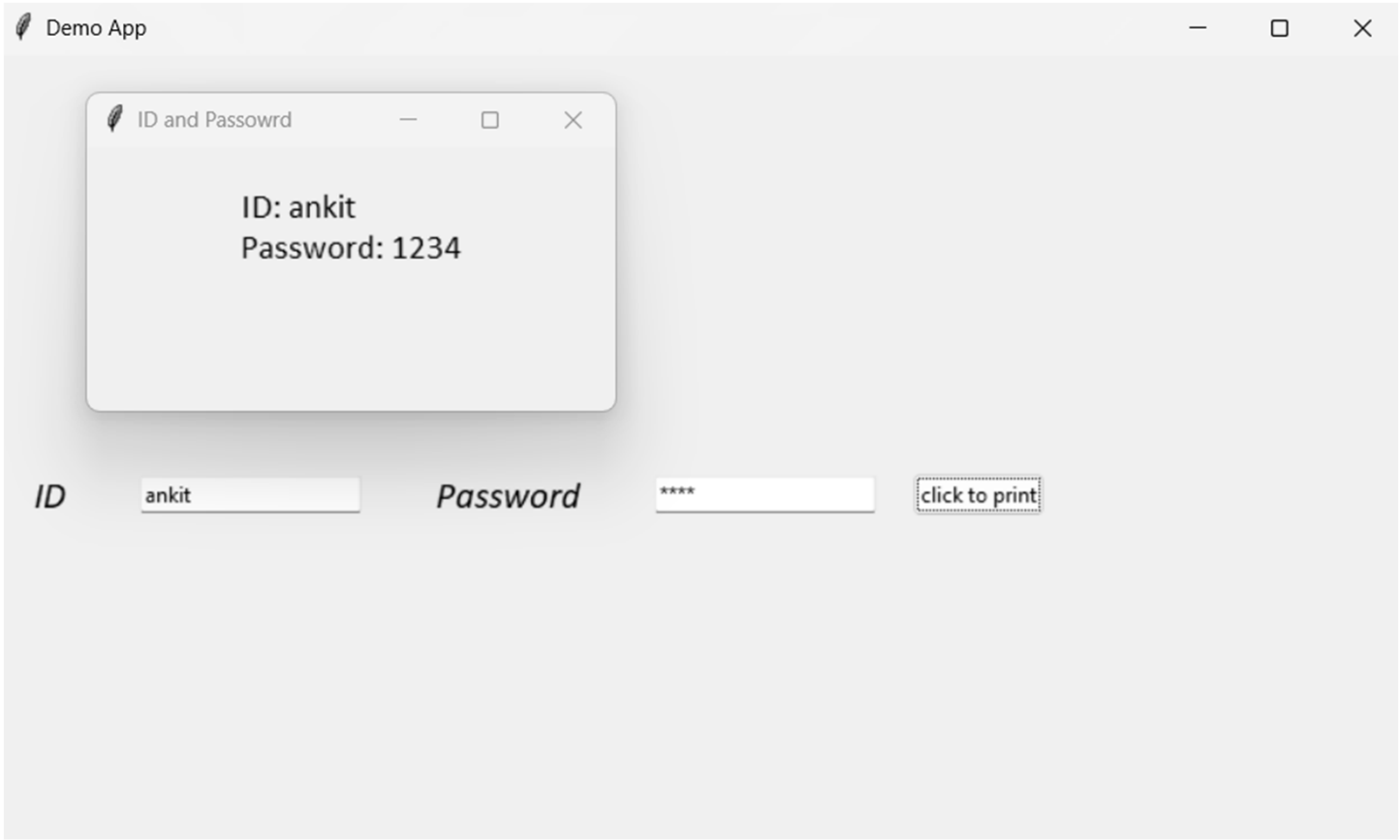
def function():
    new_window = tk.Toplevel(window)      # new window
    new_window.geometry("100x100")
    heading_1=ttk.Label(new_window,text="Hello")
    heading_1.pack()

button=ttk.Button(master=window, text="Convert", command=function)
button.pack(pady=20)

window.mainloop()
```



Assignment


Demo App
ID and Passowrd
ID: ankit
Password: 1234
ID
ankit
Password
****
click to print

Assignment

```
window = tk.Tk()
window.title("Demo App")
window.geometry("800x500") # size (Pixel)

def idpass():
    popup = tk.Toplevel(window)      # new window
    popup.title("ID and Passowrd")
    popup.geometry("300x150")

    # Get values from Entry boxes
    user_id = inputBox1.get()
    user_pass = inputBox2.get()

    label = ttk.Label(popup, text=f"ID: {user_id}\nPassword: {user_pass}", font=("Calibri", 14))
    label.pack(pady=20)

heading_1=ttk.Label(master=window,text="ID",font="calibri 16 italic")
heading_2=ttk.Label(master=window,text="Password",font="calibri 16 italic")

inputBox1=ttk.Entry(master=window)
inputBox2=ttk.Entry(master=window,show="*", )

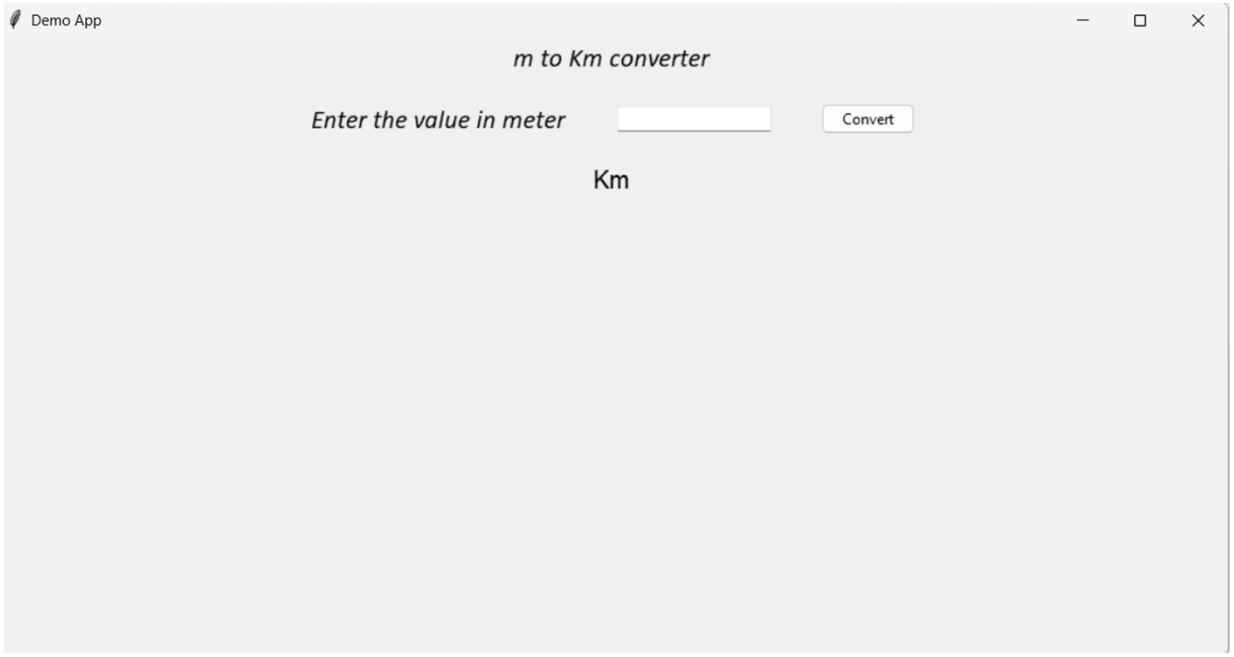
heading_1.pack(side='left',padx=20)
inputBox1.pack(side='left',padx=20)

heading_2.pack(side='left',padx=20, pady=20)
inputBox2.pack(side='left',padx=20,pady=20)

button=ttk.Button(master=window, text="click to print",command=idpass)
button.pack(side='left',pady=20)

window.mainloop()
```

Assignment


Demo App
m to Km converter
Enter the value in meter
Convert
Km

Assignment

```
import tkinter as tk # logical operator
from tkinter import ttk #widget

def mtokm():
    x=inputVar.get()
    y=x/1000
    outputVar.set(y)

window = tk.Tk()
window.title("Demo App")
window.geometry("1000x500") # size (Pixel)

heading=ttk.Label(master=window,text="m to Km converter",font="calibri 16 italic")
heading.pack()

inputVar = tk.IntVar()


inputFrame=ttk.Frame(master=window)
heading_1=ttk.Label(master=inputFrame,text="Enter the value in meter",font="calibri 16 italic")
inputBox=ttk.Entry(master=inputFrame, textvariable=inputVar)
button=ttk.Button(master=inputFrame, text="Convert",command=mtokm)


heading_1.pack(side="left",padx=20)
inputBox.pack(side="left",padx=20)
button.pack(side="left",padx=20)
inputFrame.pack(pady=20)

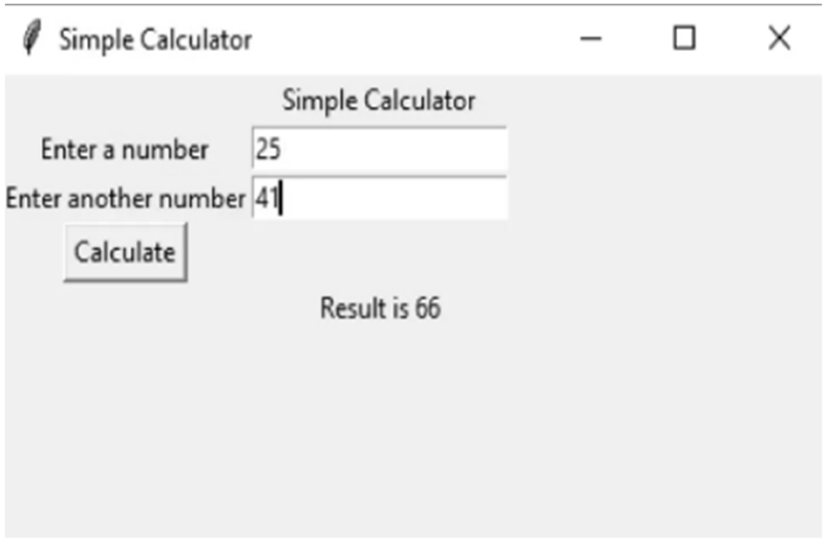
inputFrame_1=ttk.Frame(master=window)
heading_2=ttk.Label(master=inputFrame_1,text="The value in Km",font="calibri 16 italic")
heading_2.pack(side="left",padx=20)
inputFrame_1.pack(pady=20)

outputVar=tk.StringVar()
Output=ttk.Label(master=inputFrame_1,text="Km", textvariable=outputVar, font=20)
Output.pack()

window.mainloop()
```

Assignment

Develop a Python Tkinter application that takes two numbers as input, adds them, and displays the result.




```
window = tk.Tk()
window.title("Simple Calculator")
window.geometry("500x300") # size (Pixel)

def function():
    a=int(inputBox1.get())
    b=int(inputBox2.get())
    c=a+b
    heading_4=ttk.Label(master=window,text=c,font="calibri 20 bold")
    heading_5=ttk.Label(master=window,text="The result is",font="calibri 20 bold")
    heading_4.place(x=290, y=200)
    heading_5.place(x=150, y=200)

heading_1=ttk.Label(master=window,text="Simple Calculator",font="calibri 20 bold")
heading_2=ttk.Label(master=window,text="Enter a number",font="calibri 16")
heading_3=ttk.Label(master=window,text="Enter another number",font="calibri 16")

inputBox1=ttk.Entry(master=window)
inputBox2=ttk.Entry(master=window)

button=ttk.Button(master=window, text="Calculate",command=function)

heading_1.pack()

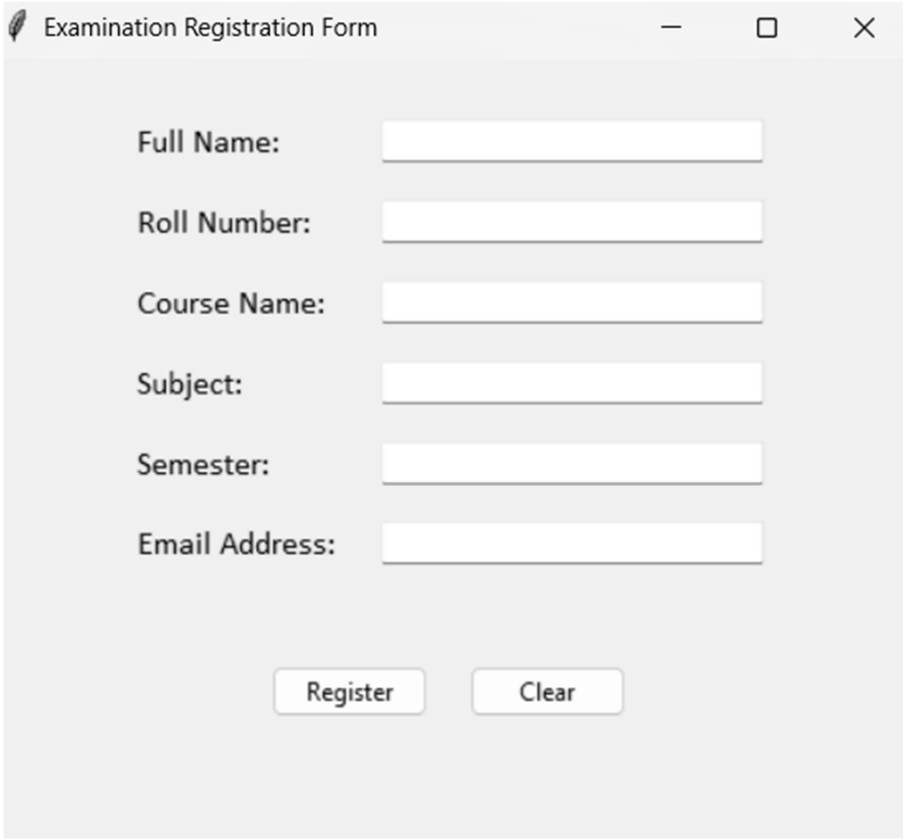
heading_2.place(x=40, y=80)
inputBox1.place(x=250, y=80)

heading_3.place(x=20, y=120)
inputBox2.place(x=250, y=120)

button.place(x=60, y=160)

window.mainloop()
```

Assignment



The GUI should include:

- Title: “Examination Registration Form”
- Labels and Entry Fields for the following details:
  - Full Name
  - Roll Number
  - Course Name
  - Subject
  - Semester
  - Email Address
- Buttons:
  - Register (to submit the form)
  - Clear (to reset all fields)

- When the Register button is clicked:
  - Validate that no field is left empty.
  - If all fields are filled:
    - Display a message box: “Registration Successful!”
  - If any field is empty:
    - Display an error message: “All fields are required!”

- When the Clear button is clicked:
  - All entry fields should be cleared/reset.

Assignment

```
import tkinter as tk
from tkinter import ttk, messagebox

# Window
window = tk.Tk()
window.title("Examination Registration Form")
window.geometry("450x400")

# Frame
form_frame = tk.Frame(window)
form_frame.pack(pady=20)

# Labels & Entry Fields
fields = ["Full Name", "Roll Number", "Course Name", "Subject", "Semester", "Email Address"]
entries = {}

for idx, field in enumerate(fields):
    label = ttk.Label(form_frame, text=field + ":", font=("Calibri", 12))
    label.grid(row=idx, column=0, padx=10, pady=8, sticky="w")

    entry = ttk.Entry(form_frame, width=30)
    entry.grid(row=idx, column=1, padx=10, pady=8)

    entries[field] = entry

# Register Button Function
def register():
    for field, entry in entries.items():
        if entry.get().strip() == "":
            messagebox.showerror("Error", "All fields are required!")
            return
    messagebox.showinfo("Success", "Registration Successful!")

# Clear Button Function
def clear_fields():
    for entry in entries.values():
        entry.delete(0, tk.END)

# Buttons
btn_frame = tk.Frame(window)
btn_frame.pack(pady=20)

register_btn = ttk.Button(btn_frame, text="Register", command=register)
register_btn.grid(row=0, column=0, padx=10)

clear_btn = ttk.Button(btn_frame, text="Clear", command=clear_fields)
clear_btn.grid(row=0, column=1, padx=10)

window.mainloop()
```